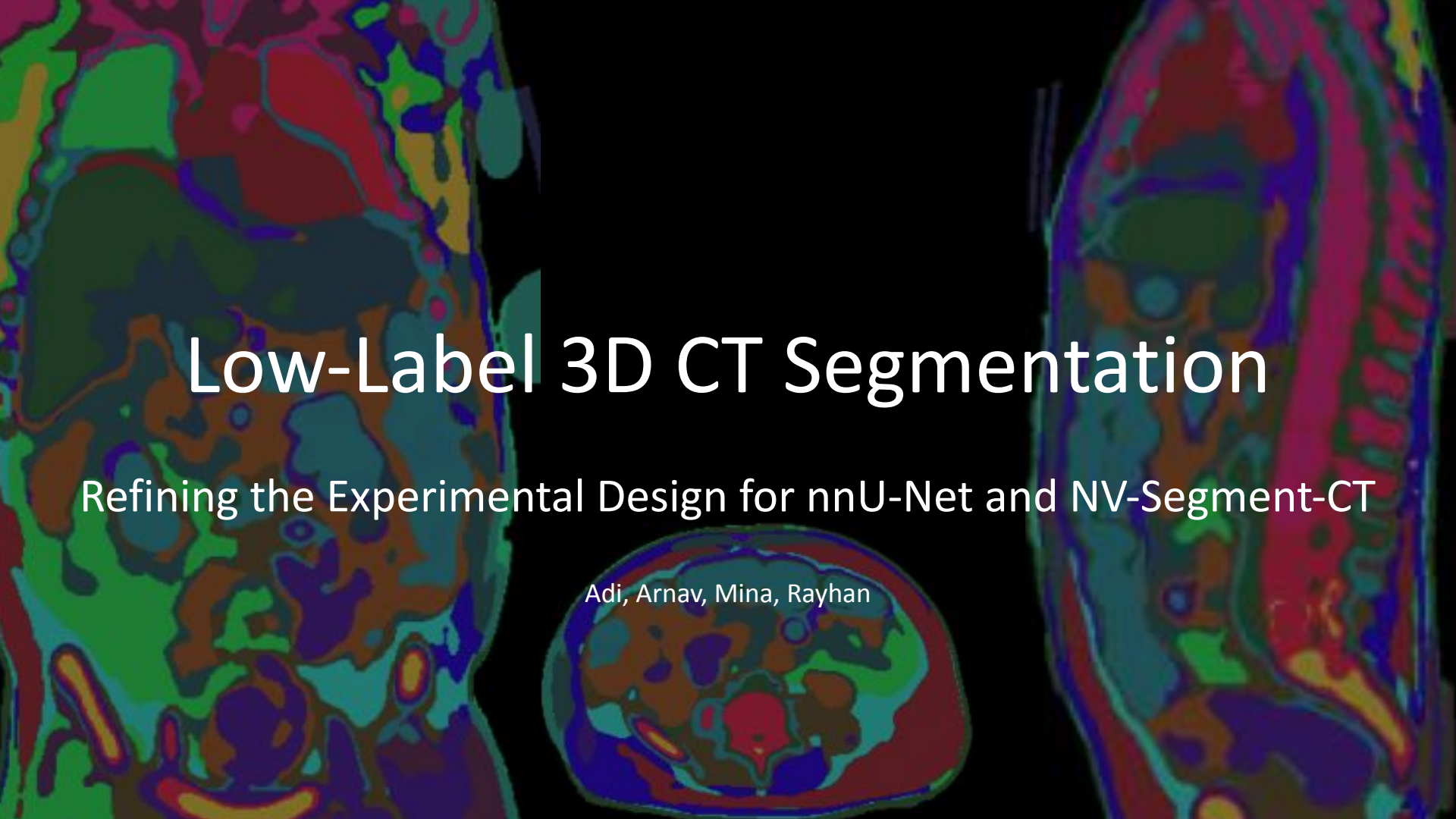


Low-Label 3D CT Segmentation

Refining the Experimental Design for nnU-Net and NV-Segment-CT

Adi, Arnav, Mina, Rayhan

Overview of Aims

Aim 1: Patient-level label scarcity	• Evaluate segmentation performance between nnU-Net and NV-Segment-CT on fully labeled patient subsets. • Segmentation masks will be benchmarked against ground-truth labels using the volumetric Dice Similarity Coefficient (DSC) for spatial overlap and the 95th-percentile Hausdorff Distance (HD95) for contour boundary fidelity
Aim 2: Distributed annotation budget	• Evaluate segmentation performance when nnU-Net and NV-Segment-CT are trained on a set of partially labeled patients • We need to take OAR size into consideration when assessing performance and adapt both models to take in partial annotations.
Aim 3: Value of pretraining	• Determine when pre-training provides an advantage over training fully from scratch • Will measure and evaluate the difference in between NV-Segment-CT and nnU-Net under every scarcity condition, and look for where pre-training helps most: at which budgets, and for which anatomical structures

Project Pipeline

Input data: SegRap2023

- 120 patients
- Paired ncCT + ceCT scans
- 45 OARs + 2 GTVs

Data preparation

- Align paired ncCT + ceCT scans
- Standardize label mappings for both NV-Segment-CT and nnU-Net

Standardization

- Standardize CT scan intensities across each input channel
- Create 2-channel 3D inputs and separate OAR + GTV training
- Analyze automatic nnU-Net preprocessing choices
- Align PAIRED ncCT + ceCT volumes

Evaluation

- Evaluate Dice for evaluating mask and ground-truth overlap
- 95th percentile hausdorff distance for boundary error

Proposal feedback - Training and Evaluation Splits

	Patient A	Patient B	Patient C	Patient D	Patient E
Fold 1	**Validate**	Train	Train	Train	Train
Fold 2	Train	**Validate**	Train	Train	Train
Fold 3	Train	Train	**Validate**	Train	Train
Fold 4	Train	Train	Train	**Validate**	Train
Fold 5	Train	Train	Train	Train	**Validate**

Label budget	Train	Validate
5	**4**	**1**
10	8	2
20	16	4
40	32	8
80	64	16

Several subsets (3 to start) of label budget's at low annotations to quantify variance

Final performance from **20 held-out** test patient samples

Proposal feedback - Model Output Descriptions

Both models will segment all 45 OARs from the SegRap2023 dataset

	nnU-Net	NV-Segment-CT
Input when testing	Paired CT scans	Paired CT Scans and 45 fixed organ IDs for class prompting
Predicted output	All 45 organs in one pass	Binary mask for each organ

We build training targets from 45 OARs

Proposal feedback - Adapting NV-Segment-CT to 45 OARs

Direct match (5)

Brain, esophagus, spinal cord, thyroid, trachea

Sub-region (10)

Brainstem, hippocampus, larynx, oral cavity, temporal lobes

Nearest: skull (18)

Cochlea, mandible, middle ear, mastoid, TMJ

No analog (14)

Chiasm, eyes, lens, optic nerves, parotids, pituitary, GTVs

Direct Match: We fine-tune the embedding directly
Sub-region: We copy the embedding from the larger region and fine-tune
Nearest: We copy the embedding from the nearest region and fine-tune
No analog: Draw a random embedding and fine-tune

We will quantify fit for organs across each of these categories

nnU-Net

Fully Annotated Patients

nnU-Net

Partially Annotated Patients

NV-Segment-CT

Fully Annotated Patients

NV-Segment-CT

Partially Annotated Patients

Aim 1A Updates: nnU-Net with Patient-Level Label Scarcity

Wrote scripts to:

1) Create patient subsets and prepare train/validation/test sets
2) Put each patient's paired CT scans and 45 organ labels into nnU-Net's expected format
3) Let nnU-Net choose preprocessing and network settings using that training subset.
4) Fit nnU-Net to the selected subset and test on the 20 patient test set
5) Compares the predicted organ masks with the expert contours and records Dice, HD95, and missed organs

Aim 1A Next Steps

1) Prepare the 5- and 10-patient subsets : draw the fixed split, convert the scans to nnU-Net's format, and run nnU-Net's preprocessing on each subset.
2) 5-epoch timing test (under 1 GPU-hour): a short run on the 5-patient subset to measure time per epoch and GPU memory. Its numbers replace our current GPU-hour estimates.
3) Train the 5- and 10-patient models (~10–15 GPU-hours each): 250 epochs each, then score on the 20 test patients with Dice and HD95. If the results look good, we plan the remaining subset sizes.

nnU-Net

Fully Annotated Patients

nnU-Net

Partially Annotated Patients

NV-Segment-CT

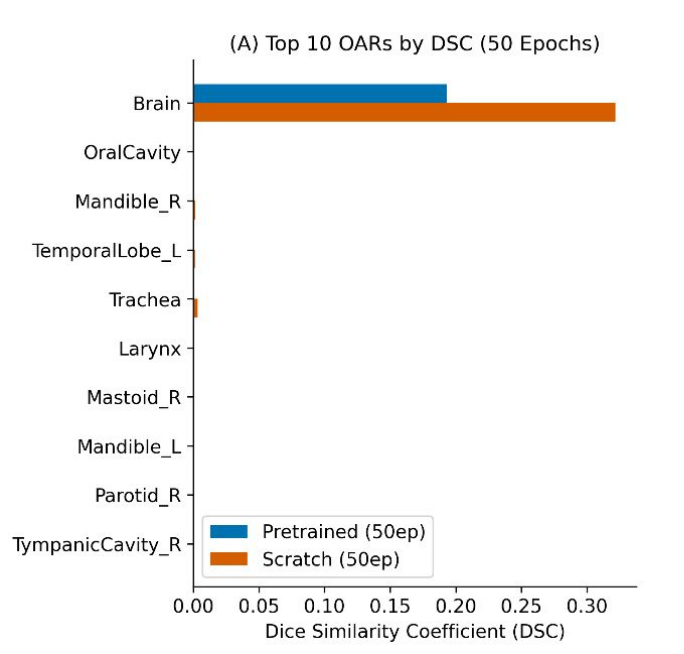
Fully Annotated Patients

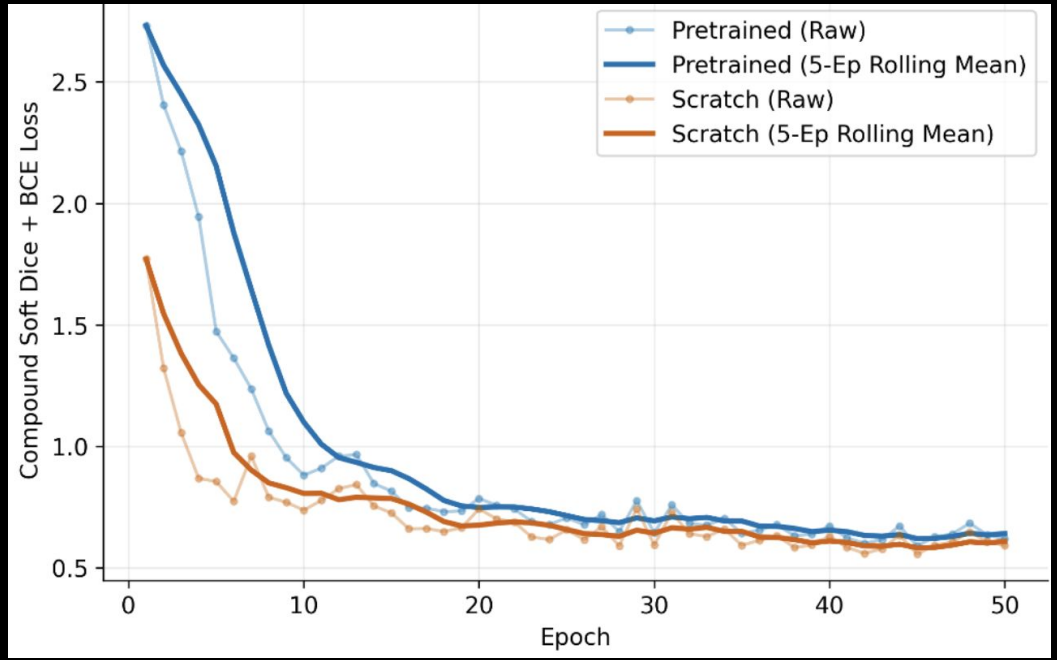
NV-Segment-CT

Partially Annotated Patients

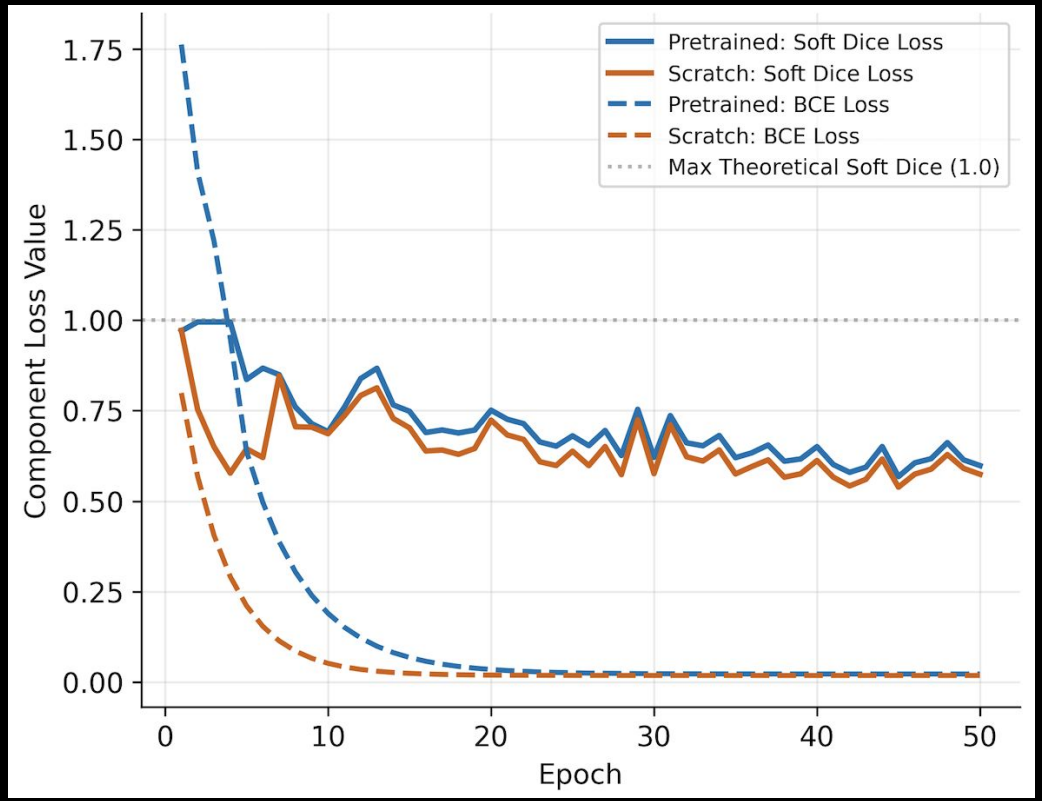
Aim 1B Updates

- Checked that Soft Dice and BCE loss are weighted equally
- BCE loss drops early on in training, but Soft Dice loss persists





Aim 1B Updates



- Changing loss weighting will probably have little impact on stopping small structures from vanishing, since Soft Dice already accounts for majority of total loss (and Soft Dice is supposed to help with smaller structures).
- Will try class-balanced foreground patch oversampling.

Aim 2 Updates

- Implemented dense and sparse annotation generator
- Updated training pipeline to support patient-specific OAR annotations and mask withheld organs
- Performed GPU and CPU smoke tests (scratch model with randomized weights) with 2 patients

```
Training on segrap_0000
Output shape: (1, 5, 32, 32, 16)
Training loss: 5.883335590362549
PASS: segrap_0000

Training on segrap_0001
Output shape: (1, 5, 32, 32, 16)
Training loss: 3.775580644607544
PASS: segrap_0001
```

Because minimum number of patients needed to test sparse annotation is 20 patients, implemented smoke test on 2 patients just to verify pipeline is working.

Aim 2B pipeline running on real patients

CPU test: data and labels

- Loads 2 real patients, both CT scans
- Crops a 32×32×16 patch around labelled organs
- Checks every withheld-organ voxel is ignored

GPU test: training step

- Runs on a V100 (32 GB), 5 organs prompted
- One training step per patient; loss is finite
- Saves a model checkpoint

GPU test

- Built the model from random weights, prompted it with 5 organs, took one training step on each patient, and saved a checkpoint
- Losses were finite, 5.88 and then 3.78
- The next step is to start from the pretrained NV-Segment-CT weights and run real training budgets